

The Challenge

Ruby is a code-breaker. She knows that the very bad people (Mr. X and Mr. Z) are sending secret messages about very bad things to each other.

However, Ruby has managed to intercept a plaintext message and the corresponding ciphertext message. By plaintext, we mean the message before it was encrypted (i.e., readable English sentences), and by ciphertext, we mean the message after it was encrypted (i.e., gibberish). To encrypt a message, each letter is changed to a new letter, so that if you read the ciphertext message, it is not obvious what the plaintext message is.

However, Ruby being the outstanding code-breaker she is, knows the algorithm that Mr. X and Mr. Z use. She knows they simply map one letter to another (possibly different) letter when they encrypt their messages. Of course, this map must be "one-to-one", meaning that each plaintext letter must correspond to exactly one ciphertext letter, as well as "onto", meaning that each ciphertext letter has exactly one corresponding plaintext letter. In this problem, there are only 27 possible characters in the plaintext and in the ciphertext - the uppercase letters and the space character.

Your job is to automate Ruby's codebreaking and help save the world.

Input

The input consists of 3 strings, with each string on a separate line. The first string is the plaintext message which Ruby knows about. The second string is the ciphertext message which corresponds to the plaintext message. The third string is another ciphertext message. You may assume that all strings have length of at least 1 character and at most 80 characters. There are only 27 possible characters in any plaintext and ciphertext: the uppercase letters (A through Z) as well as the space character ().